

```

//
=====
=====
// NAVIGATION
//
=====
=====
window.addEventListener('popstate', () => {
  if
(document.getElementById('pages').style.display === 'none') return;
  if (!history.state ||
!history.state.page)
goHomeDirectlyWithoutHistory();
});

function openSection(sectionId) {
  history.pushState({ page: sectionId },
  "");
  document.getElementById('home-
page').style.display = 'none';

document.getElementById('pages').style.display = 'block';
  document.querySelectorAll('.tool-
page').forEach(p =>
p.classList.remove('active'));

document.getElementById(sectionId).classList.add('active');

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    window.scrollTo(0, 0);
    if (sectionId === 'resume-builder-sec')
renderTemplateGrid();
    if (sectionId === 'ppt-builder-sec')
initPptBuilder();
}

function goHome() { history.back(); }

function goHomeDirectlyWithoutHistory() {

document.getElementById('pages').style.display = 'none';
    document.getElementById('home-
page').style.display = 'block';
}

//
=====
=====
// DARK MODE
//
=====
=====
function toggleDarkMode() {
    const isDark =
document.documentElement.classList.toggle('
dark');
    document.getElementById('dark-
icon').innerText = isDark ? '●' : '◐';
    document.getElementById('dark-
text').innerText = isDark ? 'Light' :
'Dark';
    try { localStorage.setItem('sh-dark-

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mode', isDark ? '1' : '0'); } catch (e) {}
}

(function initTheme() {
  let stored = null;
  try { stored = localStorage.getItem('sh-
dark-mode'); } catch (e) {}
  const prefersDark = window.matchMedia &&
window.matchMedia('(prefers-color-scheme:
dark)').matches;
  const useDark = stored === '1' || (stored
=== null && prefersDark);
  if (useDark) {

document.documentElement.classList.add('dar
k');
    document.getElementById('dark-
icon').innerText = '●';
    document.getElementById('dark-
text').innerText = 'Light';
  }
})();

//
=====
=====
// AI BACKEND CALL – talks to our own
/api/ai endpoint only.
// The Gemini key never appears in this
file or in the browser.
//
=====
=====
async function callAi(task, text) {

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    const resp = await fetch('/api/ai', {
      method: 'POST',
      headers: { 'Content-Type':
'application/json' },
      body: JSON.stringify({ task, text })
    });
    const data = await resp.json().catch(()
=> ({}));
    if (!resp.ok) {
      throw new Error(data.error || `Request
failed (${resp.status}).`);
    }
    if (!data.result) {
      throw new Error('No result returned.');
```

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//
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function updateLiveCounts() {
    const txt = document.getElementById('doc-
text').value;
    document.getElementById('live-char-
count').innerText = txt.length;
    const words =
txt.trim().split(/\s+/).filter(w =>
w.length > 0);
    document.getElementById('live-word-
count').innerText = txt.trim() ?
words.length : 0;
}

function escapeHtml(str) {
    const div =
document.createElement('div');
    div.innerText = str;
    return div.innerHTML;
}

function showDocStatus(html) {
    const el = document.getElementById('doc-
status');
    el.style.display = 'block';
    el.innerHTML = html;
}

// jsPDF and QRCode load from a CDN. On a
slow or restrictive network they can
// fail to load, which used to make
download buttons throw silently with no
```

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// feedback – the page just looked broken.
These guards check first and show
// a clear, actionable message instead of
failing silently.
function ensurePdfLib() {
  if (window.jspdf && window.jspdf.jsPDF)
    return true;
  alert("The PDF engine hasn't finished
loading yet (it loads from a CDN). Check
your connection and wait a few seconds,
then try again. If this keeps happening,
your network may be blocking
cdnjs.cloudflare.com.");
  return false;
}
function ensureDocxLib() {
  if (typeof JSZip !== 'undefined') return
true;
  alert("The Word document engine hasn't
finished loading yet (it loads from a CDN).
Check your connection and wait a few
seconds, then try again. If this keeps
happening, your network may be blocking
cdnjs.cloudflare.com.");
  return false;
}

// Offline fallback formatter – works with
no AI/network needed.
function beautifyTable() {
  const textInput =
document.getElementById('doc-text').value;
  if (!textInput.trim()) { alert("Type or
paste some text first."); return; }
```

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    const rawLines =
textInput.split('\n').filter(l =>
l.trim().length > 0);
    let subRows = "";
    rawLines.forEach(line => {
        let segment = line.trim();
        subRows += `<tr><td style="border:1px
solid
#cbd5e1;padding:10px;color:#334155;">${esca
peHtml(segment)}</td></tr>`;
    });

    window.tableFormattedOutput = `
    <h1 style="font-family:Arial,sans-
serif;color:#1e293b;margin-
bottom:14px;font-size:18pt;font-
weight:bold;">Document</h1>
    <table style="width:100%;border-
collapse:collapse;font-family:Arial,sans-
serif;">
    <tbody>${subRows}</tbody></table>`;
    showDocStatus('<span
class="ok">Done</span> – formatted as a
simple table. Download below.');
```

}

```

// AI-powered formatter – sends content to
our backend, which calls Gemini.
async function aiFormatDocument() {
    const text =
document.getElementById('doc-text').value;
    if (!text.trim()) { alert("Type or paste
some text first."); return; }
```

```
const btn = document.getElementById('ai-format-btn');
setBtnLoading(btn, 'Formatting...');
showDocStatus('Sending to AI for formatting...');

try {
  const html = await
callAi('document_format', text);
  window.tableFormattedOutput = html;
  showDocStatus('<span
class="ok">Done</span> – AI formatted your
content. Download below.');
```

 } catch (err) {

 showDocStatus(`Couldn't reach AI
formatting: \${escapeHtml(err.message)}. Try
"Format as table layout (offline)" instead,
or download as plain text.`);

 } finally {

 clearBtnLoading(btn);

 }

}

// ---- DOWNLOAD FIX ----

// Root cause of "downloads don't work" on
some mobile browsers: the old
// code called URL.revokeObjectURL(url)
immediately after a.click(), in the
// same synchronous tick. On some mobile
Chrome/WebView builds the download
// is handled asynchronously, so revoking
the blob URL before the browser
// has actually read it silently kills the
download – no error, nothing


```

// happens. Fix: revoke only after a short
delay, giving the browser time
// to start reading the blob first.
function downloadBlob(blob, filename) {
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = filename;
    a.rel = 'noopener';
    document.body.appendChild(a);
    a.click();
    document.body.removeChild(a);
    setTimeout(() =>
URL.revokeObjectURL(url), 4000);
}

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// Converts our internal simple HTML
(produced by beautifyTable / AI formatting,
// using only
h1/h2/h3/p/table/tr/td/th/strong/em/ul/li)
into an array of
// Word XML block elements (paragraphs and
tables). Falls back to treating
// the input as plain text with newlines if
no HTML structure is present.
function htmlToDocxBLOCKS(html) {
    const container =
document.createElement('div');
    container.innerHTML = html;
    const blocks = [];

    function escapeXml(s) {
        return String(s)
            .replace(/&/g, '&#x26;').replace(/</g,

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'&lt;').replace(/>/g, '&gt;');
    .replace(/"/g,
'&quot;').replace(/'/g, '&apos;');
    }

    function runsFromInline(node) {
        // Walks inline children, producing one
        <w:r> per text segment with bold/italic
        flags set.
        let runs = [];
        function walk(n, bold, italic) {
            if (n.nodeType === Node.TEXT_NODE) {
                const t = n.textContent;
                if (t) runs.push({ text: t, bold,
italic });
            }
            return;
        }
        if (n.nodeType !== Node.ELEMENT_NODE)
            return;
        const tag = n.tagName.toLowerCase();
        const nb = bold || tag === 'strong'
|| tag === 'b';
        const ni = italic || tag === 'em' ||
tag === 'i';
        n.childNodes.forEach(c => walk(c, nb,
ni));
    }
    node.childNodes.forEach(c => walk(c,
false, false));
    if (runs.length === 0) runs.push({
text: '', bold: false, italic: false });
    return runs;
}

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function runsToXml(runs) {
  return runs.map(r => {
    const props = (r.bold ? '<w:b/>' :
    '') + (r.italic ? '<w:i/>' : '');
    const rPr = props ? `<w:rPr>${props}
</w:rPr>` : '';
    return `<w:r>${rPr}<w:t
xml:space="preserve">${escapeXml(r.text)}
</w:t></w:r>`;
  }).join('');
}

function paragraphXml(node, styleId) {
  const runs = runsFromInline(node);
  const pPr = styleId ? `<w:pPr><w:pStyle
w:val="${styleId}"/></w:pPr>` : '';
  return `<w:p>${pPr}${runsToXml(runs)}
</w:p>`;
}

function tableXml(tableNode) {
  const rows =
Array.from(tableNode.querySelectorAll('tr')
);
  if (rows.length === 0) return '';
  const colCount = Math.max(...rows.map(r
=> r.children.length));
  const totalWidth = 9360; // DXA, fits
US Letter with 1" margins
  const colWidth = Math.floor(totalWidth
/ colCount);
  const gridCols = Array.from({ length:
colCount }, () => `<w:gridCol
w:w="${colWidth}"/>`).join('');

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    const rowsXml = rows.map(row => {
      const isHeaderRow = row.parentElement
      && row.parentElement.tagName.toLowerCase()
      === 'thead';
      const cells =
      Array.from(row.children).map(cell => {
        const runs = runsFromInline(cell);
        const shading = isHeaderRow ?
        '<w:shd w:val="clear" w:fill="1A2332"/>' :
        '';
        const runXml = runsToXml(runs.map(r
        => ({ ...r, bold: r.bold || isHeaderRow
        })));
        return `<w:tc><w:tcPr><w:tcW
        w:w="${colWidth}" w:type="dxa"/>${shading}
        <w:tcMar><w:top w:w="80" w:type="dxa"/>
        <w:start w:w="120" w:type="dxa"/><w:bottom
        w:w="80" w:type="dxa"/><w:end w:w="120"
        w:type="dxa"/></w:tcMar></w:tcPr>
        <w:p>${runXml}</w:p></w:tc>`;
        }).join('');
        return `<w:tr>${cells}</w:tr>`;
        }).join('');

      const borders = ['top', 'start',
      'bottom', 'end', 'insideH', 'insideV']
      .map(side => `<w:${side}
      w:val="single" w:sz="4" w:space="0"
      w:color="CBD5E1"/>`).join('');

      return `<w:tbl><w:tblPr><w:tblW
      w:w="${totalWidth}" w:type="dxa"/>
      <w:tblBorders>${borders}</w:tblBorders>

```

```

<w:tblLayout w:type="fixed"/></w:tblPr>
<w:tblGrid>${gridCols}
</w:tblGrid>${rowsXml}</w:tbl><w:p/>`;
    }

    const topNodes =
Array.from(container.childNodes).filter(n
=>
    n.nodeType === Node.ELEMENT_NODE ||
(n.nodeType === Node.TEXT_NODE &&
n.textContent.trim())
    );

    if (topNodes.length === 0) {
        // Plain text fallback: one paragraph
per line
        const text = container.textContent ||
'';
        text.split('\n').forEach(line => {
            blocks.push(`<w:p><w:r><w:t
xml:space="preserve">${escapeXml(line)}
</w:t></w:r></w:p>`);
        });
        return blocks;
    }

    topNodes.forEach(node => {
        if (node.nodeType === Node.TEXT_NODE) {
            blocks.push(`<w:p><w:r><w:t
xml:space="preserve">${escapeXml(node.textC
ontent)}</w:t></w:r></w:p>`);
            return;
        }
    })
    const tag = node.tagName.toLowerCase();

```

```

        if (tag === 'h1')
        blocks.push(paragraphXml(node,
        'Heading1'));
        else if (tag === 'h2')
        blocks.push(paragraphXml(node,
        'Heading2'));
        else if (tag === 'h3')
        blocks.push(paragraphXml(node,
        'Heading3'));
        else if (tag === 'table')
        blocks.push(tableXml(node));
        else if (tag === 'ul' || tag === 'ol')
        {
            Array.from(node.children).forEach(li
            => {
                const runs = runsFromInline(li);
                blocks.push(`

```

```

async function buildDocxBlob(bodyXmlBlocks)
{
    const zip = new JSZip();

    zip.file('[Content_Types].xml',
        `<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<Types
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/content-types">
    <Default Extension="rels"
ContentType="application/vnd.openxmlformats
-package.relationships+xml"/>
    <Default Extension="xml"
ContentType="application/xml"/>
    <Override PartName="/word/document.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.wordprocessingml.document.ma
in+xml"/>
    <Override PartName="/word/styles.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.wordprocessingml.styles+xml"
/>
    <Override PartName="/word/numbering.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.wordprocessingml.numbering+x
ml"/>
</Types>`);

    zip.file('_rels/.rels',
        `<?xml version="1.0" encoding="UTF-8"

```

```

standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/officeDocume
nt" Target="word/document.xml"/>
</Relationships>`);

```

```

zip.file('word/_rels/document.xml.rels',
`<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/styles"
Target="styles.xml"/>
  <Relationship Id="rId2"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/numbering"
Target="numbering.xml"/>
</Relationships>`);

```

```

zip.file('word/numbering.xml',
`<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<w:numbering
xmlns:w="http://schemas.openxmlformats.org/
wordprocessingml/2006/main">
  <w:abstractNum w:abstractNumId="0">
    <w:lvl w:ilvl="0"><w:numFmt

```



```

w:val="bullet"/><w:lvlText
w:val="&#8226;"/><w:lvlJc w:val="left"/>
    <w:pPr><w:ind w:left="720"
w:hanging="360"/></w:pPr></w:lvl>
    </w:abstractNum>
    <w:num w:numId="1"><w:abstractNumId
w:val="0"/></w:num>
</w:numbering>`);

zip.file('word/styles.xml',
    `<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<w:styles
xmlns:w="http://schemas.openxmlformats.org/
wordprocessingml/2006/main">
    <w:docDefaults>
        <w:rPrDefault><w:rPr><w:rFonts
w:ascii="Arial" w:hAnsi="Arial"/><w:sz
w:val="22"/></w:rPr></w:rPrDefault>
    </w:docDefaults>
    <w:style w:type="paragraph" w:default="1"
w:styleId="Normal">
        <w:name w:val="Normal"/><w:pPr>
<w:spacing w:after="160" w:line="288"
w:lineRule="auto"/></w:pPr>
    </w:style>
    <w:style w:type="paragraph"
w:styleId="Heading1">
        <w:name w:val="heading 1"/><w:basedOn
w:val="Normal"/><w:next w:val="Normal"/>
        <w:pPr><w:spacing w:before="240"
w:after="160"/></w:pPr>
        <w:rPr><w:b/><w:sz w:val="32"/><w:color
w:val="0B1120"/></w:rPr>

```

```

</w:style>
<w:style w:type="paragraph"
w:styleId="Heading2">
  <w:name w:val="heading 2"/><w:basedOn
w:val="Normal"/><w:next w:val="Normal"/>
  <w:pPr><w:spacing w:before="200"
w:after="120"/></w:pPr>
  <w:rPr><w:b/><w:sz w:val="27"/><w:color
w:val="0B1120"/></w:rPr>
</w:style>
<w:style w:type="paragraph"
w:styleId="Heading3">
  <w:name w:val="heading 3"/><w:basedOn
w:val="Normal"/><w:next w:val="Normal"/>
  <w:pPr><w:spacing w:before="160"
w:after="100"/></w:pPr>
  <w:rPr><w:b/><w:sz w:val="24"/><w:color
w:val="0B1120"/></w:rPr>
</w:style>
</w:styles>`);

```

```

const bodyXml = bodyXmlBlocks.join('');
zip.file('word/document.xml',
  `<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<w:document
xmlns:w="http://schemas.openxmlformats.org/
wordprocessingml/2006/main">
  <w:body>
    ${bodyXml}
  <w:sectPr>
    <w:pgSz w:w="12240" w:h="15840"/>
    <w:pgMar w:top="1440" w:right="1440"
w:bottom="1440" w:left="1440"

```

```

w:header="720" w:footer="720"
w:gutter="0"/>
    </w:sectPr>
</w:body>
</w:document>`);

    return zip.generateAsync({
        type: 'blob',
        mimeType:
'application/vnd.openxmlformats-
officedocument.wordprocessingml.document'
    });
}

async function downloadWord() {
    const text =
document.getElementById('doc-text').value;
    if (!text.trim() &&
!window.tableFormattedOutput) { alert("Type
something first."); return; }
    if (!ensureDocxLib()) return;

    const btn = document.querySelector('#doc-
pdf-maker .btn-primary');
    const originalLabel = btn ? btn.innerText
: null;
    if (btn) { btn.disabled = true;
btn.innerText = 'Building .docx...'; }

    try {
        const sourceHtml =
window.tableFormattedOutput ||
text.split('\n').map(line => `<p>${line}</p>`).join('');

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        const blocks =
htmlToDocxBLOCKS(sourceHtml);
        const blob = await
buildDocxBlob(blocks);
        downloadBlob(blob, 'Document.docx');
        showDocStatus('<span
class="ok">Download started</span> –
Document.docx should open in Word, Google
Docs, or any mobile Office app.');
```

```

    } catch (err) {
        console.error('docx build error:',
err);
        showDocStatus("Couldn't build the Word
document. Please try again, or use Download
.pdf instead.");
    } finally {
        if (btn) { btn.disabled = false;
btn.innerText = originalLabel; }
    }
}

function downloadPdf() {
    const text =
document.getElementById('doc-text').value;
    if (!text.trim() &&
!window.tableFormattedOutput) { alert("Type
something first."); return; }
    if (!ensurePdfLib()) return;
    const { jsPDF } = window.jspdf;
    const doc = new jsPDF();
    const stripped =
window.tableFormattedOutput
?
window.tableFormattedOutput.replace(/<\tr>

```

```

/g, '\n').replace(/<\td><td[^>]*>/g, ' |
').replace(/<[^>]*>/g, '')
    : text;
    const lines =
doc.splitTextToSize(stripped, 180);
    doc.text(lines, 15, 20);
    doc.save('Document.pdf');
    showDocStatus('<span class="ok">Download
started</span> – check your
downloads/notifications.');
```

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// 5. IMAGES TO PDF
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function convertImagesToPdf() {
    const files =
Array.from(document.getElementById('img-to-
pdf-input').files);
    const status =
document.getElementById('img-to-pdf-
status');
    if (files.length === 0) { alert("Choose
at least one image first."); return; }
    if (!ensurePdfLib()) return;
    const btn = document.getElementById('img-
to-pdf-btn');
    btn.disabled = true; btn.innerText =
"Converting...";
    status.style.display = 'block';
```

```

    status.innerHTML = `Reading <span
class="num">${files.length}</span>
image(s)...`;

    const readFile = (file) => new
Promise((resolve, reject) => {
    const reader = new FileReader();
    reader.onload = (e) =>
resolve(e.target.result);
    reader.onerror = reject;
    reader.readAsDataURL(file);
});

    Promise.all(files.map(readFile))
    .then((dataUrls) => {
        const { jsPDF } = window.jspdf;
        const doc = new jsPDF();
        dataUrls.forEach((dataUrl, i) => {
            if (i > 0) doc.addPage();
            doc.addImage(dataUrl, 'JPEG', 10,
10, 190, 270);
        });
        doc.save('Images.pdf');
        status.innerHTML = `<span
class="ok">Done</span> – ${dataUrls.length}
page(s) saved to Images.pdf`;
    })
    .catch(() => { status.innerHTML =
`Couldn't read one of the images. Try
again.`; })
    .finally(() => { btn.disabled = false;
btn.innerText = "Convert & download PDF";
});
}

```

```

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// 1. PHOTO COMPRESSOR
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=====
const RESIZE_PRESETS = {
  passport_2x2in:    { w: 600, h: 600 },
  passport_35x45mm: { w: 413, h: 531 },
  stamp_20x25mm:    { w: 236, h: 295 },
  a4_300dpi:         { w: 2480, h: 3508 },
};

let originalImgDims = null; // {w, h} of
the currently loaded image, for aspect-lock
math

function previewOriginalImage() {
  const file =
document.getElementById('photo-
input').files[0];
  const out =
document.getElementById('original-size');
  originalImgDims = null;
  if (!file) { out.style.display = 'none';
return; }

  out.style.display = 'block';
  out.innerHTML = `Original size: <span
class="num">${(file.size /
1024).toFixed(1)} KB</span>`;

```

```
const reader = new FileReader();
reader.onload = (e) => {
  const img = new Image();
  img.onload = () => {
    originalImgDims = { w: img.width, h:
img.height };
    out.innerHTML += ` &nbsp;&nbsp;  `;
<span class="num">${img.width}
×${img.height}</span> px`;
    if (document.getElementById('custom-
width').value === '')
document.getElementById('custom-
width').value = img.width;
    if (document.getElementById('custom-
height').value === '')
document.getElementById('custom-
height').value = img.height;
  };
  img.src = e.target.result;
};
reader.readAsDataURL(file);
}

function onResizePresetChange() {
  const preset =
document.getElementById('resize-
preset').value;
  const customRow =
document.getElementById('custom-dim-row');
  const aspectRow =
document.getElementById('aspect-lock-row');
  if (preset === 'custom') {
    customRow.style.display = 'grid';
    aspectRow.style.display = 'flex';
  }
}
```



```

    } else {
        customRow.style.display = 'none';
        aspectRow.style.display = 'none';
    }
}

function onCustomDimInput(changed) {
    const lock =
document.getElementById('aspect-
lock').checked;
    if (!lock || !originalImgDims) return;
    const ratio = originalImgDims.w /
originalImgDims.h;
    const widthEl =
document.getElementById('custom-width');
    const heightEl =
document.getElementById('custom-height');
    if (changed === 'width') {
        const w = parseFloat(widthEl.value);
        if (w > 0) heightEl.value =
Math.round(w / ratio);
    } else {
        const h = parseFloat(heightEl.value);
        if (h > 0) widthEl.value = Math.round(h
* ratio);
    }
}

function getTargetDimensions(naturalW,
naturalH) {
    const preset =
document.getElementById('resize-
preset').value;
    if (preset === 'original') return { w:

```

```

naturalW, h: naturalH };
    if (preset === 'custom') {
        const w =
parseInt(document.getElementById('custom-
width').value, 10);
        const h =
parseInt(document.getElementById('custom-
height').value, 10);
        if (w > 0 && h > 0) return { w, h };
        return { w: naturalW, h: naturalH };
    }
    const p = RESIZE_PRESETS[preset];
    return p ? { w: p.w, h: p.h } : { w:
naturalW, h: naturalH };
}

function processAdvancedPhoto() {
    const input =
document.getElementById('photo-
input').files[0];
    if (!input) { alert("Choose an image
first."); return; }
    const targetKb =
parseFloat(document.getElementById('target-
kb').value);
    const format =
document.getElementById('photo-
format').value;
    const btn =
document.getElementById('compress-btn');
    btn.disabled = true; btn.innerText =
"Compressing...";

    const reader = new FileReader();

```

```

    reader.onload = (e) => {
      const img = new Image();
      img.onload = () => {
        const { w, h } =
getTargetDimensions(img.width, img.height);
        const canvas =
document.createElement('canvas');
        const ctx = canvas.getContext('2d');
        canvas.width = w; canvas.height = h;
        ctx.drawImage(img, 0, 0, w, h);

        let q = 0.85;
        let dataUrl =
canvas.toDataURL(format, q);
        if (targetKb && format ===
'image/jpeg') {
          let min = 0.01, max = 0.98;
          for (let i = 0; i < 10; i++) {
            q = (min + max) / 2;
            dataUrl =
canvas.toDataURL(format, q);
            const sizeKb =
(dataUrl.split(',')[1].length * 0.75) /
1024;
            if (sizeKb > targetKb) max = q;
          else min = q;
          }
        }
        const a =
document.createElement('a');
        a.href = dataUrl; a.download =
'optimized.' + (format === 'image/png' ?
'png' : 'jpg');
        document.body.appendChild(a);

```

```

        a.click();
        document.body.removeChild(a);
        btn.disabled = false; btn.innerText =
"Compress & download";
    };
    img.onerror = () => {
        alert("Couldn't read that image. Try
a different file.");
        btn.disabled = false; btn.innerText =
"Compress & download";
    };
    img.src = e.target.result;

};
reader.onerror = () => {
    alert("Couldn't read that file.");
    btn.disabled = false; btn.innerText =
"Compress & download";
};
reader.readAsDataURL(input);
}

//
=====
=====
// 8. AGE CALCULATOR
//
=====
=====
function calculateAge() {
    const dobValue =
document.getElementById('dob-input').value;
    const out = document.getElementById('age-
result');

```

```

    if (!dobValue) { alert("Pick a date of
    birth first."); return; }
    const dob = new Date(dobValue);
    if (isNaN(dob)) return;
    if (dob > new Date()) { alert("That date
    is in the future."); return; }

    const now = new Date();
    let years = now.getFullYear() -
    dob.getFullYear();
    let months = now.getMonth() -
    dob.getMonth();
    let days = now.getDate() - dob.getDate();
    if (days < 0) { months -= 1; days += new
    Date(now.getFullYear(), now.getMonth(),
    0).getDate(); }
    if (months < 0) { years -= 1; months +=
    12; }

    out.style.display = 'block';
    out.innerHTML = `

```

```
const RESUME_TEMPLATES = [
  { id: 'professional', name:
'Professional', swatch: 'linear-
gradient(135deg,#100E1A,#332E4D)' },
  { id: 'creative', name:
'Creative', swatch: 'linear-
gradient(135deg,#FF6B35,#E0A030)' },
  { id: 'minimal', name:
'Minimal', swatch: '#FBF8F2' },
  { id: 'modern-teal', name: 'Modern
Teal', swatch: 'linear-
gradient(135deg,#0FA3A3,#0B1120)' },
  { id: 'violet-side', name: 'Violet
Side', swatch: 'linear-
gradient(135deg,#7C5CFC,#332E4D)' },
  { id: 'classic-line', name: 'Classic
Line', swatch: 'linear-gradient(180deg,#fff
60%,#100E1A 60%)' },
  { id: 'two-tone', name: 'Two-
Tone', swatch: 'linear-
gradient(90deg,#100E1A 35%,#FBF8F2 35%)' },
  { id: 'bold-header', name: 'Bold
Header', swatch: 'linear-
gradient(180deg,#E14B4B 30%,#fff 30%)' },
  { id: 'compact-grid', name: 'Compact
Grid', swatch: 'repeating-linear-
gradient(90deg,#E7E0D2 0 2px,#fff 2px
20px)' },
  { id: 'elegant-serif',name: 'Elegant
Serif',swatch: 'linear-
gradient(135deg,#332E4D,#9892AE)' },
];

let dynamicTemplate = 'professional';
```

```

function renderTemplateGrid() {
  const grid =
document.getElementById('tpl-grid');
  if (grid.dataset.rendered) return;
  grid.innerHTML = RESUME_TEMPLATES.map(t
=> `
    <div class="tpl-card ${t.id} ===
dynamicTemplate ? 'is-active' : ''" data-
tpl="${t.id}"
onclick="setResumeTemplate('${t.id}')">
      <div class="tpl-swatch"
style="background:${t.swatch};"></div>
      <span>${t.name}</span>
    </div>
  `).join('');
  grid.dataset.rendered = '1';
}

function setResumeTemplate(id) {
  dynamicTemplate = id;
  document.querySelectorAll('.tpl-
card').forEach(c => {
    c.classList.toggle('is-active',
c.dataset.tpl === id);
  });
}

function downloadResume() {
  if (!ensurePdfLib()) return;
  const name =
document.getElementById('res-name').value
|| "Your Name";
  const email =

```

```
document.getElementById('res-email').value
|| "your.email@example.com";
    const phone =
document.getElementById('res-phone').value;
    const edu = document.getElementById('res-
education').value || "Add your education
details";
    const experience =
document.getElementById('res-
experience').value;
    const skills =
document.getElementById('res-skills').value
|| "Add your skills";

    const { jsPDF } = window.jspdf;
    const doc = new jsPDF();
    const pageWidth = 210;
    let margin = 20;
    let y = 0;

    function section(title, body, startY, x,
width, color) {
        doc.setFont(undefined, 'bold');
doc.setFontSize(11);
        if (color) doc.setTextColor(...color);
else doc.setTextColor(20, 20, 20);
        doc.text(title.toUpperCase(), x,
startY);
        doc.setFont(undefined, 'normal');
doc.setFontSize(10.5);
        doc.setTextColor(20, 20, 20);
        const lines = doc.splitTextToSize(body,
width);
        doc.text(lines, x, startY + 6);
```



```

    return startY + 6 + lines.length * 5.2
+ 8;
    }

    const contactLine = [email,
phone].filter(Boolean).join(' · ');

    switch (dynamicTemplate) {
      case 'professional': {
        doc.setFillColor(16, 14, 26);
doc.rect(0, 0, pageWidth, 38, 'F');
        doc.setTextColor(255, 255, 255);
doc.setFontSize(20); doc.setFont(undefined,
'bold');
        doc.text(name, 20, 20);
        doc.setFontSize(10);
doc.setFont(undefined, 'normal');
doc.text(contactLine, 20, 29);
        y = 52; margin = 20;
        y = section('Education', edu, y,
margin, 172);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 172);
        section('Skills', skills, y, margin,
172);
        break;
      }
      case 'creative': {
        doc.setFillColor(255, 107, 53);
doc.rect(0, 0, 50, 297, 'F');
        doc.setTextColor(255, 255, 255);
doc.setFontSize(15); doc.setFont(undefined,
'bold');

```

```

        const nameLines =
doc.splitTextToSize(name, 38);
        doc.text(nameLines, 8, 24);
        doc.setFontSize(9);
doc.setFont(undefined, 'normal');
        doc.text(doc.splitTextToSize([email,
phone].filter(Boolean).join('\n'), 38), 8,
24 + nameLines.length * 7 + 6);
        margin = 65; y = 25;
        y = section('Education', edu, y,
margin, 125);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 125);
        section('Skills', skills, y, margin,
125);
        break;
    }
    case 'minimal': {
        doc.setFontSize(22);
doc.setFont(undefined, 'bold');
doc.setTextColor(20,20,20); doc.text(name,
20, 22);
        doc.setLineWidth(0.4); doc.line(20,
27, pageWidth - 18, 27);
        doc.setFontSize(10);
doc.setFont(undefined, 'normal');
doc.text(contactLine, 20, 34);
        y = 46; margin = 20;
        y = section('Education', edu, y,
margin, 172);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 172);

```

```

        section('Skills', skills, y, margin,
172);
        break;
    }
    case 'modern-teal': {
        doc.setFillColor(15, 163, 163);
doc.rect(0, 0, pageWidth, 44, 'F');
        doc.setTextColor(255,255,255);
doc.setFontSize(22);
doc.setFont(undefined,'bold');
doc.text(name, 20, 24);
        doc.setFontSize(10);
doc.setFont(undefined,'normal');
doc.text(contactLine, 20, 34);
        y = 58; margin = 20;
        y = section('Education', edu, y,
margin, 172, [15,163,163]);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 172, [15,163,163]);
        section('Skills', skills, y, margin,
172, [15,163,163]);
        break;
    }
    case 'violet-side': {
        doc.setFillColor(124, 92, 252);
doc.rect(0, 0, 60, 297, 'F');
        doc.setTextColor(255,255,255);
doc.setFontSize(16);
doc.setFont(undefined,'bold');
        const nl = doc.splitTextToSize(name,
48); doc.text(nl, 8, 26);
        doc.setFontSize(9);
doc.setFont(undefined,'normal');

```

```

        doc.text(doc.splitTextToSize([email,
phone].filter(Boolean).join('\n'), 48), 8,
26 + nl.length*7 + 8);
        margin = 70; y = 28;
        y = section('Education', edu, y,
margin, 120, [124,92,252]);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 120, [124,92,252]);
        section('Skills', skills, y, margin,
120, [124,92,252]);
        break;
    }
    case 'classic-line': {
        doc.setFontSize(24);
doc.setFont(undefined, 'bold');
doc.setTextColor(20,20,20);
        doc.text(name, pageWidth/2, 26, {
align: 'center' });
        doc.setFontSize(10);
doc.setFont(undefined, 'normal');
        doc.text(contactLine, pageWidth/2,
34, { align: 'center' });
        doc.setLineWidth(0.6); doc.line(20,
40, pageWidth-20, 40);
        y = 52; margin = 20;
        y = section('Education', edu, y,
margin, 172);
        if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 172);
        section('Skills', skills, y, margin,
172);
        break;
    }
}

```

```

    }
    case 'two-tone': {
        doc.setFillStyle(16,14,26);
doc.rect(0, 0, 74, 297, 'F');
        doc.setTextColor(255,255,255);
doc.setFontSize(15);
doc.setFont(undefined,'bold');
        const nl2 = doc.splitTextToSize(name,
60); doc.text(nl2, 8, 24);
        doc.setFontSize(9);
doc.setFont(undefined,'normal');
        doc.text(doc.splitTextToSize([email,
phone].filter(Boolean).join('\n'), 60), 8,
24 + nl2.length*7 + 8);
        doc.setFontSize(10);
doc.setFont(undefined,'bold');
doc.text('SKILLS', 8, 100);
        doc.setFont(undefined,'normal');
doc.setFontSize(9);
        doc.text(doc.splitTextToSize(skills,
60), 8, 107);
        margin = 84; y = 25;
        y = section('Education', edu, y,
margin, 108);
        if (experience.trim())
section('Experience / Projects',
experience, y, margin, 108);
        break;
    }
    case 'bold-header': {
        doc.setFillStyle(225, 75, 75);
doc.rect(0, 0, pageWidth, 50, 'F');
        doc.setTextColor(255,255,255);
doc.setFontSize(24);

```

```

doc.setFont(undefined,'bold');
doc.text(name, 20, 28);
    doc.setFontSize(10);
doc.setFont(undefined,'normal');
doc.text(contactLine, 20, 40);
    y = 64; margin = 20;
    y = section('Education', edu, y,
margin, 172, [225,75,75]);
    if (experience.trim()) y =
section('Experience / Projects',
experience, y, margin, 172, [225,75,75]);
        section('Skills', skills, y, margin,
172, [225,75,75]);
        break;
    }
    case 'compact-grid': {
        doc.setFontSize(18);
doc.setFont(undefined,'bold');
doc.setTextColor(20,20,20); doc.text(name,
20, 20);
        doc.setFontSize(9);
doc.setFont(undefined,'normal');
doc.text(contactLine, 20, 27);
        doc.setDrawColor(231,224,210);
doc.setLineWidth(0.3);
        doc.line(20, 32, pageWidth-20, 32);
        y = 42; margin = 20;
        y = section('Education', edu, y,
margin, 80);
        const rightY = section('Skills',
skills, 42, 110, 80);
        if (experience.trim())
section('Experience / Projects',
experience, Math.max(y, rightY), margin,

```

```

172);
    break;
}
case 'elegant-serif':
default: {
    doc.setFont('times', 'bold');
doc.setFontSize(22);
doc.setTextColor(20,20,20); doc.text(name,
20, 24);
    doc.setFont('times', 'normal');
doc.setFontSize(11); doc.text(contactLine,
20, 32);
    doc.setLineWidth(0.3); doc.line(20,
37, pageWidth-20, 37);
    doc.setFont('times', 'bold');
doc.setFontSize(11); doc.text('EDUCATION',
20, 48);
    doc.setFont('times', 'normal');
doc.setFontSize(10.5);
    let eduLines =
doc.splitTextToSize(edu, 172);
doc.text(eduLines, 20, 54);
    let yy = 54 + eduLines.length*5.5 +
8;
    if (experience.trim()) {
        doc.setFont('times','bold');
doc.text('EXPERIENCE / PROJECTS', 20, yy);
        doc.setFont('times','normal');
        let explLines =
doc.splitTextToSize(experience, 172);
doc.text(explLines, 20, yy+6);
        yy = yy + 6 + explLines.length*5.5 +
8;
    }
}

```

```

        doc.setFont('times','bold');
doc.text('SKILLS', 20, yy);
        doc.setFont('times','normal');
doc.text(doc.splitTextToSize(skills, 172),
20, yy+6);
        break;
    }
}

doc.save('Resume.pdf');
}

//
=====
=====
// 10. GRAMMAR CHECKER – offline quick
check + AI fix-it
//
=====
=====
function auditContent() {
    const text =
document.getElementById('grammar-check-
text').value;
    const out =
document.getElementById('audit-output');
    document.getElementById('ai-fix-
output').classList.add('is-hidden');
    out.classList.remove('is-hidden');

    if (!text.trim()) { out.innerText =
"Paste some text first."; return; }

    const issues = [];

```



```

    if (/ +/.test(text)) issues.push("Double
spaces found – check for accidental extra
spacing.");
    const repeatMatch =
text.match(/\b(\w+)\s+\1\b/i);
    if (repeatMatch) issues.push(`Repeated
word found: "${repeatMatch[1]}
${repeatMatch[1]}".`);
    if (/[a-z]\.[A-Z]/.test(text))
issues.push("Missing space after a period
between sentences.");
    if (/ \bi\b/.test(text))
issues.push('Lowercase "i" found – the
pronoun "I" should always be
capitalized.');
```

```

    if (!/[.!?] \s*$/.test(text.trim()))
issues.push("The text doesn't end with a
period, question mark, or exclamation
mark.");
```

```

    const sentences =
text.split(/[.!?]+/).filter(s =>
s.trim().length > 0);
    const longSentences = sentences.filter(s
=> s.trim().split(/\s+/).length > 35);
    if (longSentences.length > 0)
issues.push(`${longSentences.length}
sentence(s) are quite long (35+ words) –
consider splitting them up.`);
```

```

    const commonTypos = { 'teh': 'the',
'recieve': 'receive', 'seperate':
'separate', 'occured': 'occurred',
'definately': 'definitely', 'alot': 'a lot'
```

```

};
Object.keys(commonTypos).forEach( typo =>
{
    const re = new RegExp(`\\b${typo}\\b`,
    'i');
    if (re.test(text))
    issues.push(`Possible typo: "${typo}"
    &rarr; did you mean "${commonTypos[typo]}"?
    `);
    });

    const words =
    text.trim().split(/\s+/).filter(w =>
    w.length > 0);
    const summary = `Checked ${words.length}
    words across ${sentences.length || 1}
    sentence(s).`;

    out.innerHTML = issues.length > 0
    ? `

"${summary}"</p><p style="margin:0 0
    6px;">Found ${issues.length} potential
    issue(s):</p><ul style="margin:0;padding-
    left:18px;">${issues.map(i => `"${summary}"</p><p style="margin:0;">No
    common issues detected by these checks.
    Still worth a careful read-through, or try
    "Find & fix mistakes" for a deeper AI pass.
    </p>`;
}

async function aiFixGrammar() {


```

```

    const text =
document.getElementById('grammar-check-
text').value;
    const out = document.getElementById('ai-
fix-output');
    document.getElementById('audit-
output').classList.add('is-hidden');
    out.classList.remove('is-hidden');

    if (!text.trim()) { out.innerText =
"Paste some text first."; return; }

    const btn = document.getElementById('ai-
fix-btn');
    setBtnLoading(btn, 'Checking...');
    out.innerHTML = 'Sending text to AI for
review...';

    try {
        const raw = await callAi('grammar_fix',
text);
        const splitIdx =
raw.indexOf('CHANGES:');
        const corrected = (splitIdx >= 0 ?
raw.slice(0, splitIdx) : raw).trim();
        const changes = splitIdx >= 0 ?
raw.slice(splitIdx + 8).trim() : '';

        out.innerHTML = `
            <p style="font-weight:600;margin:0 0
8px;">Corrected text:</p>
            <p style="white-space:pre-
wrap;background:var(--paper-deep);border-
radius:8px;padding:10px;margin:0 0

```

```

10px;">${escapeHtml(corrected)}</p>
    <button class="btn btn-ghost"
style="width:auto;display:inline-
flex;padding:8px 14px;font-
size:12px;margin-bottom:10px;"
onclick="applyAiFix()">Use this corrected
text</button>
    ${changes ? `<p style="font-
weight:600;margin:10px 0 6px;">What was
changed:</p><p style="white-space:pre-
wrap;">${escapeHtml(changes)}</p>` : ''}
    `;
    out.dataset.corrected = corrected;
  } catch (err) {
    out.innerHTML = `Couldn't reach the AI
service: ${escapeHtml(err.message)}. You
can still try the offline quick check.`;
  } finally {
    clearBtnLoading(btn);
  }
}

function applyAiFix() {
  const out = document.getElementById('ai-
fix-output');
  if (out.dataset.corrected) {
    document.getElementById('grammar-check-
text').value = out.dataset.corrected;
  }
}

//
=====
=====

```

```

// 11. POWERPOINT GENERATOR – offline
outline -> .pptx
// Builds a real OOXML PowerPoint package
by hand using JSZip
// (same approach as the Word builder
above). No AI, no network.
//
=====

// Five color themes. Hex values are 6-char
RRGGBB (no '#'), as required
// inside <a:srgbClr val="...">.
const PPT_THEMES = [
  { id: 'midnight', name: 'Midnight',
    bg: '100E1A', accent: 'FF6B35', text:
    'FFFFFF', sub: 'C9C4DA', swatch: 'linear-
    gradient(135deg,#100E1A,#332E4D)' },
  { id: 'sunrise', name: 'Sunrise',
    bg: 'FBF8F2', accent: 'FF6B35', text:
    '100E1A', sub: '6E6884', swatch: 'linear-
    gradient(135deg,#FBF8F2,#FFE4D6)' },
  { id: 'teal-deep', name: 'Teal Deep',
    bg: '0B1120', accent: '0FA3A3', text:
    'FFFFFF', sub: 'B7D9D9', swatch: 'linear-
    gradient(135deg,#0FA3A3,#0B1120)' },
  { id: 'violet', name: 'Violet',
    bg: '1B1830', accent: '7C5CFC', text:
    'FFFFFF', sub: 'C9C4DA', swatch: 'linear-
    gradient(135deg,#7C5CFC,#1B1830)' },
  { id: 'classic', name: 'Classic
    White', bg: 'FFFFFF', accent: 'E14B4B',
    text: '1A1A1A', sub: '595959', swatch:
    'linear-gradient(180deg,#fff 60%,#E14B4B

```

```

60%)' },
];

let pptTheme = 'midnight';
let pptGridRendered = false;

function initPptBuilder() {
  renderPptThemeGrid();
  const outline =
document.getElementById('ppt-outline');
  if (!outline.dataset.bound) {
    outline.addEventListener('input',
updatePptSlideCount);
    outline.dataset.bound = '1';
  }
  updatePptSlideCount();
}

function renderPptThemeGrid() {
  const grid =
document.getElementById('ppt-tpl-grid');
  if (pptGridRendered) return;
  grid.innerHTML = PPT_THEMES.map(t => `
    <div class="tpl-card ${t.id} ===
pptTheme ? 'is-active' : ''" data-
tpl="${t.id}"
onclick="setPptTheme('${t.id}')">
      <div class="tpl-swatch"
style="background:${t.swatch};"></div>
      <span>${t.name}</span>
    </div>
  `).join('');
  grid.querySelectorAll =
grid.querySelectorAll; // no-op, keeps

```

```

structure consistent
  pptGridRendered = true;
}

function setPptTheme(id) {
  pptTheme = id;
  document.querySelectorAll('#ppt-tpl-grid
.tpl-card').forEach(c => {
    c.classList.toggle('is-active',
c.dataset.tpl === id);
  });
}

// Splits the outline textarea into slides.
Blank lines separate slides;
// within a block, the first line is the
slide title and the rest become
// bullet points.
function parsePptOutline(raw) {
  const blocks = raw.split(/\n\s*\n/).map(b
=> b.trim()).filter(Boolean);
  return blocks.map(block => {
    const lines = block.split('\n').map(l
=> l.trim()).filter(Boolean);
    const title = (lines[0] || 'Untitled
slide').replace(/:$/, '');
    const bullets = lines.slice(1);
    return { title, bullets };
  });
}

function updatePptSlideCount() {
  const raw = document.getElementById('ppt-
outline').value;

```

```

    const slides = parsePptOutline(raw);
    const el = document.getElementById('ppt-
slide-count');
    el.innerText = `${slides.length} slide(s)
will be generated (plus a title slide).`;
}

function xmlEscape(str) {
    return String(str)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&apos;');
}

// Builds a single <p:sp> text box XML
fragment.
function pptTextBox(opts) {
    const { x, y, cx, cy, text, sizePt, bold,
color, align, bullets } = opts;
    const alignAttr = align ? `
align="${align}"` : '';
    let paragraphs;
    if (bullets && bullets.length) {
        paragraphs = bullets.map(line => `
        <a:p>
            <a:pPr marL="285750"
indent="-285750"${alignAttr}>
                <a:buFont typeface="Arial"/>
                <a:buChar char="#8226;"/>
            </a:pPr>
            <a:r>
                <a:rPr lang="en-US" sz="${sizePt

```



```

* 100}" dirty="0">
    <a:solidFill><a:srgbClr
val="${color}"/></a:solidFill>
    </a:rPr>
    <a:t>${xmlEscape(line)}</a:t>
</a:r>
</a:p>`).join('');
} else {
    paragraphs = `
    <a:p>
    <a:pPr${alignAttr}/>
    <a:r>
    <a:rPr lang="en-US" sz="${sizePt
* 100}" ${bold ? 'b="1"' : ''} dirty="0">
    <a:solidFill><a:srgbClr
val="${color}"/></a:solidFill>
    </a:rPr>
    <a:t>${xmlEscape(text)}</a:t>
    </a:r>
    </a:p>`;
}
return `
    <p:sp>
    <p:nvSpPr>
    <p:cNvPr id="0" name="TextBox"/>
    <p:cNvSpPr txBox="1"/>
    <p:nvPr/>
    </p:nvSpPr>
    <p:spPr>
    <a:xfrm><a:off x="${x}" y="${y}"/>
    <a:ext cx="${cx}" cy="${cy}"/></a:xfrm>
    <a:prstGeom prst="rect"><a:avLst/>
    </a:prstGeom>
    <a:noFill/>

```

```

    </p:spPr>
    <p:txBody>
        <a:bodyPr wrap="square" anchor="t">
<a:noAutofit/></a:bodyPr>
        <a:lstStyle/>
        ${paragraphs}
    </p:txBody>
</p:sp>`;
}

// Full-bleed background rectangle in the
// theme color.
function pptBackgroundRect(bg, EMU_W,
EMU_H) {
    return `
    <p:sp>
        <p:nvSpPr><p:cNvPr id="0"
name="Background"/><p:cNvSpPr/><p:nvPr/>
</p:nvSpPr>
        <p:spPr>
            <a:xfrm><a:off x="0" y="0"/><a:ext
cx="${EMU_W}" cy="${EMU_H}"/></a:xfrm>
            <a:prstGeom prst="rect"><a:avLst/>
</a:prstGeom>
            <a:solidFill><a:srgbClr
val="${bg}"/></a:solidFill>
            <a:ln><a:noFill/></a:ln>
        </p:spPr>
        <p:txBody><a:bodyPr/><a:lstStyle/>
<a:p/></p:txBody>
    </p:sp>`;
}

// Thin accent bar used on title + content

```

```

slides for visual identity.
function pptAccentBar(accent, x, y, cx, cy)
{
    return `
        <p:sp>
            <p:nvSpPr><p:cNvPr id="0"
name="AccentBar"/><p:cNvSpPr/><p:nvPr/>
</p:nvSpPr>
            <p:spPr>
                <a:xfrm><a:off x="${x}" y="${y}"/>
<a:ext cx="${cx}" cy="${cy}"/></a:xfrm>
                <a:prstGeom prst="rect"><a:avLst/>
</a:prstGeom>
                <a:solidFill><a:srgbClr
val="${accent}"/></a:solidFill>
                <a:ln><a:noFill/></a:ln>
            </p:spPr>
            <p:txBody><a:bodyPr/><a:lstStyle/>
<a:p/></p:txBody>
        </p:sp>`;
}

function buildTitleSlideXml(theme, title,
subtitle, EMU_W, EMU_H) {
    const shapes = [
        pptBackgroundRect(theme.bg, EMU_W,
EMU_H),
        pptAccentBar(theme.accent, 838200,
2667000, 1219200, 45720),
        pptTextBox({ x: 838200, y: 2743200, cx:
10515600, cy: 1219200, text: title, sizePt:
40, bold: true, color: theme.text }),
    ];
    if (subtitle && subtitle.trim()) {

```

```

        shapes.push(pptTextBox({ x: 838200, y:
3962400, cx: 10515600, cy: 609600, text:
subtitle, sizePt: 18, bold: false, color:
theme.sub }));
    }
    return shapes.join('');
}

function buildContentSlideXml(theme, slide,
EMU_W, EMU_H) {
    const shapes = [
        pptBackgroundRect(theme.bg, EMU_W,
EMU_H),
        pptAccentBar(theme.accent, 838200,
838200, 914400, 45720),
        pptTextBox({ x: 838200, y: 990600, cx:
10515600, cy: 762000, text: slide.title,
sizePt: 28, bold: true, color: theme.text
}),
    ];
    if (slide.bullets.length) {
        shapes.push(pptTextBox({
            x: 838200, y: 1981200, cx: 10515600,
cy: 4267200,
            bullets: slide.bullets, sizePt: 18,
color: theme.text,
        }));
    }
    return shapes.join('');
}

function pptSlideXmlWrapper(innerShapes) {
    return `<?xml version="1.0"
encoding="UTF-8" standalone="yes"?>

```

```

<p:sld
xmlns:a="http://schemas.openxmlformats.org/
drawingml/2006/main"
xmlns:r="http://schemas.openxmlformats.org/
officeDocument/2006/relationships"
xmlns:p="http://schemas.openxmlformats.org/
presentationml/2006/main">
  <p:cSld>
    <p:spTree>
      <p:nvGrpSpPr><p:cNvPr id="1"
name=""/><p:cNvGrpSpPr/><p:nvPr/>
</p:nvGrpSpPr>
      <p:grpSpPr><a:xfrm><a:off x="0"
y="0"/><a:ext cx="0" cy="0"/><a:chOff x="0"
y="0"/><a:chExt cx="0" cy="0"/></a:xfrm>
</p:grpSpPr>
      ${innerShapes}
    </p:spTree>
  </p:cSld>
  <p:clrMapOvr><a:masterClrMapping/>
</p:clrMapOvr>
</p:sld>`;
}

```

```

async function buildPptxBlob(title,
subtitle, slides, themeId) {
  const theme = PPT_THEMES.find(t => t.id
=== themeId) || PPT_THEMES[0];
  const EMU_W = 12192000; // 16:9, 13.333in
  const EMU_H = 6858000; // 7.5in
  const zip = new JSZip();

  zip.file('[Content_Types].xml', `<?xml
version="1.0" encoding="UTF-8"

```

```

standalone="yes"?>
<Types
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/content-types">
  <Default Extension="rels"
ContentType="application/vnd.openxmlformats
-package.relationships+xml"/>
  <Default Extension="xml"
ContentType="application/xml"/>
  <Override
PartName="/ppt/presentation.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.presentationml.presentation.
main+xml"/>
  <Override
PartName="/ppt/slideMasters/slideMaster1.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.presentationml.slideMaster+xml"/>
  <Override
PartName="/ppt/slideLayouts/slideLayout1.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.presentationml.slideLayout+xml"/>
  <Override
PartName="/ppt/theme/theme1.xml"
ContentType="application/vnd.openxmlformats
-officedocument.theme+xml"/>
  ${slides.map((_, i) => `<Override

```

```

PartName="/ppt/slides/slide${i + 1}.xml"
ContentType="application/vnd.openxmlformats
-
officedocument.presentationml.slide+xml"/>`
).join('\n ')}
</Types>`);

    zip.file('_rels/.rels', `<?xml
version="1.0" encoding="UTF-8"
standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/officeDocume
nt" Target="ppt/presentation.xml"/>
</Relationships>`);

    const slideRelEntries = slides.map((_, i)
=>
      `<Relationship Id="rId${i + 2}"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/slide"
Target="slides/slide${i + 1}.xml"/>`
    ).join('\n ');

    zip.file('ppt/_rels/presentation.xml.rels',
`<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">

```

```

    <Relationship Id="rId1"
    Type="http://schemas.openxmlformats.org/off
    iceDocument/2006/relationships/slideMaster"
    Target="slideMasters/slideMaster1.xml"/>
    ${slideRelEntries}
    <Relationship Id="rId99"
    Type="http://schemas.openxmlformats.org/off
    iceDocument/2006/relationships/theme"
    Target="theme/theme1.xml"/>
  </Relationships>`);

```

```

    const sldIdLst = slides.map((_, i) =>
    `<p:sldId id="${256 + i}" r:id="rId${i +
    2}"/>`).join('');

```

```

    zip.file('ppt/presentation.xml', `<?xml
    version="1.0" encoding="UTF-8"
    standalone="yes"?>
    <p:presentation
    xmlns:a="http://schemas.openxmlformats.org/
    drawingml/2006/main"
    xmlns:r="http://schemas.openxmlformats.org/
    officeDocument/2006/relationships"
    xmlns:p="http://schemas.openxmlformats.org/
    presentationml/2006/main">
      <p:sldMasterIdLst><p:sldMasterId
    id="2147483648" r:id="rId1"/>
    </p:sldMasterIdLst>
      <p:sldIdLst>${sldIdLst}</p:sldIdLst>
      <p:sldSz cx="${EMU_W}" cy="${EMU_H}"/>
      <p:notesSz cx="6858000" cy="9144000"/>
    </p:presentation>`);

```

```

    zip.file('ppt/theme/theme1.xml', `<?xml

```



```
version="1.0" encoding="UTF-8"
standalone="yes"?>
<a:theme
xmlns:a="http://schemas.openxmlformats.org/
drawingml/2006/main"
name="StudentHubTheme">
  <a:themeElements>
    <a:clrScheme name="StudentHub">
      <a:dk1><a:srgbClr val="000000"/>
    </a:dk1>
      <a:lt1><a:srgbClr val="FFFFFF"/>
    </a:lt1>
      <a:dk2><a:srgbClr val="{theme.bg}"/>
    </a:dk2>
      <a:lt2><a:srgbClr
val="{theme.text}"/></a:lt2>
      <a:accent1><a:srgbClr
val="{theme.accent}"/></a:accent1>
      <a:accent2><a:srgbClr
val="{theme.accent}"/></a:accent2>
      <a:accent3><a:srgbClr
val="{theme.accent}"/></a:accent3>
      <a:accent4><a:srgbClr
val="{theme.accent}"/></a:accent4>
      <a:accent5><a:srgbClr
val="{theme.accent}"/></a:accent5>
      <a:accent6><a:srgbClr
val="{theme.accent}"/></a:accent6>
      <a:hlink><a:srgbClr
val="{theme.accent}"/></a:hlink>
      <a:folHlink><a:srgbClr
val="{theme.accent}"/></a:folHlink>
    </a:clrScheme>
    <a:fontScheme name="StudentHub">
```

```
<a:majorFont><a:latin
typeface="Arial"/></a:majorFont>
  <a:minorFont><a:latin
typeface="Arial"/></a:minorFont>
</a:fontScheme>
  <a:fmtScheme name="StudentHub">
    <a:fillStyleLst>
      <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
      <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
      <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
    </a:fillStyleLst>
    <a:lnStyleLst>
      <a:ln w="6350"><a:solidFill>
<a:schemeClr val="accent1"/></a:solidFill>
</a:ln>
      <a:ln w="12700"><a:solidFill>
<a:schemeClr val="accent1"/></a:solidFill>
</a:ln>
      <a:ln w="19050"><a:solidFill>
<a:schemeClr val="accent1"/></a:solidFill>
</a:ln>
    </a:lnStyleLst>
    <a:effectStyleLst>
      <a:effectStyle><a:effectLst/>
</a:effectStyle>
      <a:effectStyle><a:effectLst/>
</a:effectStyle>
      <a:effectStyle><a:effectLst/>
</a:effectStyle>
    </a:effectStyleLst>
    <a:bgFillStyleLst>
```

```

        <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
        <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
        <a:solidFill><a:schemeClr
val="accent1"/></a:solidFill>
    </a:bgFillStyleLst>
</a:fmtScheme>
</a:themeElements>
</a:theme>`);

```

```

zip.file('ppt/slideMasters/slideMaster1.xml
', `<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<p:sldMaster
xmlns:a="http://schemas.openxmlformats.org/
drawingml/2006/main"
xmlns:r="http://schemas.openxmlformats.org/
officeDocument/2006/relationships"
xmlns:p="http://schemas.openxmlformats.org/
presentationml/2006/main">
    <p:cSld>
        <p:bg><p:bgRef idx="1001"><a:schemeClr
val="bg1"/></p:bgRef></p:bg>
        <p:spTree>
            <p:nvGrpSpPr><p:cNvPr id="1"
name=""><p:cNvGrpSpPr><p:nvPr/>
</p:nvGrpSpPr>
            <p:grpSpPr/>
        </p:spTree>
    </p:cSld>
    <p:clrMap bg1="lt1" tx1="dk1" bg2="lt2"
tx2="dk2" accent1="accent1"

```

```

accent2="accent2" accent3="accent3"
accent4="accent4" accent5="accent5"
accent6="accent6" hlink="hlink"
folHlink="folHlink"/>
  <p:sldLayoutIdLst><p:sldLayoutId
id="2147483649" r:id="rId1"/>
</p:sldLayoutIdLst>
</p:sldMaster>`);

```

```

zip.file('ppt/slideMasters/_rels/slideMaste
r1.xml.rels', `<?xml version="1.0"
encoding="UTF-8" standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/slideLayout"
Target="../../slideLayouts/slideLayout1.xml"/>
</Relationships>`);

```

```

zip.file('ppt/slideLayouts/slideLayout1.xml
', `<?xml version="1.0" encoding="UTF-8"
standalone="yes"?>
<p:sldLayout
xmlns:a="http://schemas.openxmlformats.org/
drawingml/2006/main"
xmlns:r="http://schemas.openxmlformats.org/
officeDocument/2006/relationships"
xmlns:p="http://schemas.openxmlformats.org/
presentationml/2006/main" type="blank"
preserve="1">

```

```

<p:cSld name="Blank">
  <p:spTree>
    <p:nvGrpSpPr><p:cNvPr id="1"
name=""/><p:cNvGrpSpPr/><p:nvPr/>
  </p:nvGrpSpPr>
    <p:grpSpPr/>
  </p:spTree>
</p:cSld>
  <p:clrMapOvr><a:masterClrMapping/>
</p:clrMapOvr>
</p:sldLayout>`);

```

```

zip.file('ppt/slidelayouts/_rels/slideLayout1.xml.rels', `<?xml version="1.0"
encoding="UTF-8" standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/package/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/slideMaster"
Target="../slideMasters/slideMaster1.xml"/>
</Relationships>`);

```

```

slides.forEach((slide, i) => {
  const innerShapes = i === 0
    ? buildTitleSlideXml(theme, title,
      subtitle, EMU_W, EMU_H)
    : buildContentSlideXml(theme, slide,
      EMU_W, EMU_H);
  zip.file(`ppt/slides/slide${i +
1}.xml`, pptSlideXmlWrapper(innerShapes));
  zip.file(`ppt/slides/_rels/slide${i +

```

```

1}.xml.rels`, `<?xml version="1.0"
encoding="UTF-8" standalone="yes"?>
<Relationships
xmlns="http://schemas.openxmlformats.org/pa
ckage/2006/relationships">
  <Relationship Id="rId1"
Type="http://schemas.openxmlformats.org/off
iceDocument/2006/relationships/slideLayout"
Target="../../slideLayouts/slideLayout1.xml"/>
</Relationships>`);
});

return zip.generateAsync({
  type: 'blob',
  mimeType:
'application/vnd.openxmlformats-
officedocument.presentationml.presentation'
,
});
}

async function downloadPpt() {
  const title =
document.getElementById('ppt-
title').value.trim() || 'Untitled
Presentation';
  const subtitle =
document.getElementById('ppt-
subtitle').value.trim();
  const outlineRaw =
document.getElementById('ppt-
outline').value;
  const contentSlides =
parsePptOutline(outlineRaw);

```

```
    if (!contentSlides.length) {
      alert("Add at least one slide to your
outline first (a title line, optionally
followed by bullet points).");
      return;
    }
    if (!ensureDocxLib()) return; // JSZip
guard – same lib powers both .docx and
.pptx

    const btn = document.getElementById('ppt-
download-btn');
    setBtnLoading(btn, 'Building .pptx...');

    try {
      // Slide 1 is always the title slide;
the rest come from the outline.
      const allSlides = [{ title, bullets: []
}, ...contentSlides];
      const blob = await buildPptxBlob(title,
subtitle, allSlides, pptTheme);
      const safeName = title.replace(/^[a-z0-
9\_ ]/gi, '').trim().replace(/\s+/g, '\_')
|| 'Presentation';
      downloadBlob(blob, `${safeName}.pptx`);
    } catch (err) {
      console.error('pptx build error:',
err);
      alert("Couldn't build the PowerPoint
file. Please try again – if it keeps
happening, try simplifying the outline.");
    } finally {
      clearBtnLoading(btn);
    }
  }
}
```

```
}  
}
```